

# Roshan Agrawal

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## Education

**Birla Institute of Technology and Science Pilani**  
*Bachelor of Engineering, Electrical and Electronics Engineering*

Pilani, Rajasthan  
May 2024

## Experience

### Software Development Engineer II

July 2024 – Present

*Tata Communications*

*Bengaluru, India*

#### *Multi-threaded Logger*

- Architected project in **Java SpringBoot** for centralized **audit logging** microservice on **event-driven** architecture to ingest, order, and persist customer lifecycle events across **100+ Kubernetes** clusters spanning multiple datacenters
- Scaled concurrent **RabbitMQ** consumer ingestion by developing **multi-threaded algorithm** using **BlockingQueue** and **ConcurrentHashMap** for asynchronous ingestion, increasing **throughput 5x**
- Enforced sequential consistency under distributed infrastructure conditions using **background thread scheduling** with a **10s SLA**, compensating for **network latency** and persisted data in Postgres

#### *AI Studio & Data Platform*

- Operated within AI Studio's managed LLM MLOps stack (Vector DB, Workspaces, ML services, Model Hub, Fine-tuning, Guardrails) and Data Platform's big data infrastructure (Airflow, Doris, Clickhouse, Kafka)
- Built the **backend provisioning** layer for Vector Databases- **Milvus** and **Qdrant** on Kubernetes of a **3-tier architecture**, for enterprise **RAG** applications, exposing RESTful APIs across a multi-cluster setup contributing to revenue of **\$2 Million** of the platform
- Benchmarked** and tuned **Clickhouse** against AWS **Redshift** from **1GB to 50TB scale**, beating **Redshift** by **60%** in ingestion and **10x** in query time **winning JLR** as a customer
- Built end-to-end **observability and monitoring stack** for the Vector DB platform, exposing metrics via **Prometheus**, wrote Alertmanager rules, stress tested the **alerts** and designed **Grafana** dashboards cutting MTTD from **30 min to 10 min**
- Packaged and containerized microservices on Docker and deployed on Kubernetes across multi-cluster cloud setups, managing workloads via Helm charts and custom resource configurations
- Wrote unit tests using JUnit, configured CI/CD pipelines using Github Actions, led Playwright UI testing for Angular frontend, used GenAI tools, documented vector DB features and addressed customer issues on-call

### Intern

July 2023 – June 2024

*Uber*

*Hyderabad, India*

- Audited 40+ CRM metrics across **BigQuery datawarehouse** agent ticket tables, designed **ER diagram** in LucidChart, validated upstream and downstream data coherency, and consolidated a production SSOT table improving team **efficiency by 13%**
- Developed agent ticket **BI datapipeline** using **Python and SQL** with **cron scheduling**, validated via **YAML config**, and deployed to production, saving 1200 man hours and **cutting costs by \$12K**
- Built CSAT dashboard in **Tableau**, **automated** weekly SQL to GSheets **workflows** via uWorc using Python scripting, and conducted deepdives **production datasets**; insights presented to director informing **\$15M/month** in business decisions

## Projects

### **Startup Project - Carbon Capture** | *Genetic Engineering, Microbial Ecosystem, Biotechnology*

- Engineered a **microbial ecosystem** for **carbon capture**, achieving **95% accuracy** and **80% cost reduction**
- Won awards from **Avery Dennison** and **Falling Wall Labs** and received an interview call from **Antler India**

### **Drone Monitoring System** | *NodeMCU, RaspberryPi, MQTT, Python, Firebase, Bootstrap, Django*

- Developed a drone monitoring system integrating IoT hardware and a full-stack Django application

### **Spotify web-app** | *HTML, CSS, TypeScript, React.js, Node.js, Express.js, MongoDB, REST APIs*

- Created a full-stack web application integrating Spotify API with a RESTful backend

## Skills

- Tech-stack and Framework:** Java, C++, Kubernetes, Spring Boot, Spring Data JPA, MVC, RabbitMQ, Docker, SQL, C, HTML, CSS, Postgres, Redis, Python, Golang, NoSQL database, Kafka, Hibernate, Maven
- Tools and Skills:** Git, Postman, Linux, APIs, Github Copilot, Cursor, Claude Code, ChatGPT, AWS
- CS Fundamentals:** Backend development, DevOps, Distributed Systems, Data Structures and Algorithms, Object Oriented Programming, Multithreading, Database Management Systems, Computer networks, System Design, Operating Systems